Martin Orozco Jr.

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Objective

Junior Electrical Engineering student pursuing a Summer 2027 ASIC/VLSI Hardware Engineering internship in RTL design and functional verification. Designs and verifies Verilog RTL (FIFOs, a pipelined RV32I RISC-V CPU with hazard detection) using self-checking testbenches, waveform-based debug, and FPGA validation.

Education

Bachelor of Science, Electrical Engineering — Expected Dec 2027
San José State University — San José, CA — GPA: 3.75

- Relevant Coursework: Digital Logic Design, Signals and Systems, Electronic Circuits, Embedded Systems Design (ARM, C, Assembly) — in progress

Electrical Engineering Transfer Degree — May 2025
Mission College — Santa Clara, CA — GPA: 3.75

Technical Skills

Digital Design / Verification: RTL Design, Functional Verification, Self-Checking Testbenches, Waveform Debug, FPGA Implementation, Digital Logic Design
Languages: Verilog, Python, C++, C, MATLAB
Tools: Icarus Verilog, GTKWave, Xilinx Vivado, Git/GitHub, Altium Designer (Schematic Capture), LTspice, Oscilloscopes, Logic Analyzers

RTL / Digital Design Projects

Portfolio: github.com/m-orozco/chip-design-portfolio — *UART, AXI-Lite, and a pipelined RV32I RISC-V CPU (hazard detection, FPGA target) in progress*

Synchronous FIFO Buffer — Verilog, Icarus Verilog, GTKWave: Designed a parameterizable synchronous FIFO using an extra-MSB read/write pointer scheme to disambiguate full vs. empty without a separate counter. Built a self-checking testbench validating data integrity, overflow guarding, underflow guarding, and write/read order preservation, with all assertions passing and cross-verified against GTKWave waveforms.

FPGA RTL Digital Arithmetic Modules — Verilog, Vivado, Nexys A7: Implemented and verified a majority logic unit, 1-bit full adder, and hierarchical 4-bit ripple-carry adder; validated functionality via simulation and on-hardware FPGA testing.

Python-Based Circuit Design Automation: Developed Python scripts to automate RLC filter design calculations and support validation of circuit performance, improving design efficiency and reducing manual errors.

Experience

SJSU Formula SAE Spartan Racing — Oct 2025 – May 2026
Electrical Engineering Intern, Sensor & Harness Team — San José, CA

- **Sensor Integration & Validation**: Validated vehicle sensor systems by verifying wiring connections, electrical interfaces, and signal functionality.
- **Harness Testing & Debug**: Performed continuity checks and structured debug on sensor harnesses to isolate wiring faults and confirm reliable connections.
- **Low Voltage BMS**: Supported LVBMS PCB design and worked with sensor communication interfaces, including I2C.

BMW of Fremont — Nov 2011 – Aug 2025
Foreman / Master ASE Technician — Fremont, CA

- **CAN Bus Diagnostics**: Diagnosed CAN bus network integrity and signal-level faults using an oscilloscope and BMW ISTA diagnostic software across vehicle electrical systems.
- **Deep Root-Cause Diagnosis**: Served as final escalation point for complex, hard-to-diagnose electrical faults, tracing issues to root cause to reduce repeat comebacks.
- **Technical Leadership**: Led a team of 24 technicians as Foreman, enforcing standardized diagnostic procedures and consistent quality across the shop.